

Final model · Stage 5: the final test

Sealed 2015 H2 test split opened once · frozen engine v1.0.0 · criteria committed beforehand (commit 8c5b9ab)

Result on the sealed test set (2015 H2, 212,801 loans, opened once): ROC-AUC **0.7465** (validation 0.7351), mean PD 19.6% vs actual 20.1%. The engine ranks risk better than **LendingClub's own interest rate** (ROC-AUC 0.7148, +0.032). **7 of 8** pre-registered criteria met; the miss is the known 60-month calibration gap.

A. How the test was run

The test loans were never used for any choice: not for training, early stopping, calibration, thresholds or the zip-code decision. They were scored with the frozen bundle (checksums verified on load). As a final end-to-end proof, 5,000 test loans were also scored by the deployable engine from their **original raw CSV rows**: maximum PD difference **0.0**.

B. Test vs validation

Metric	Validation (2015 H1)	TEST (2015 H2)	Change
Default rate	20.4%	20.1%	-0.29 pts
Mean predicted PD	20.4%	19.6%	-0.72 pts
Calibration gap (PD – actual)	0.0%	-0.4%	-0.43 pts
ROC-AUC	0.7351	0.7465	+0.0114
PR-AUC	0.4195	0.4336	+0.0141
KS	0.3404	0.3597	+0.0193
Brier score (lower = better)	0.1424	0.1388	-0.0036
Log loss (lower = better)	0.4458	0.4361	-0.0097
Expected calibration error	0.3%	0.6%	+0.25 pts

Test scores are slightly *better* than validation. That is not leakage: the test months simply ranked more easily. LendingClub's own interest rate improves by the same amount (ROC-AUC 0.7037 on validation → 0.7148 on test, +0.011, vs our +0.011). The engine kept its lead over LendingClub in both periods.

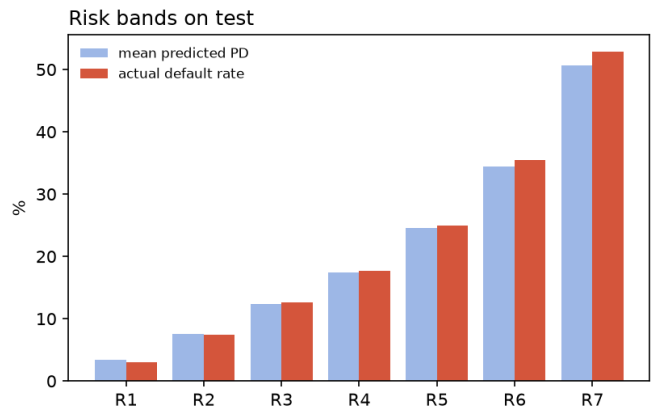
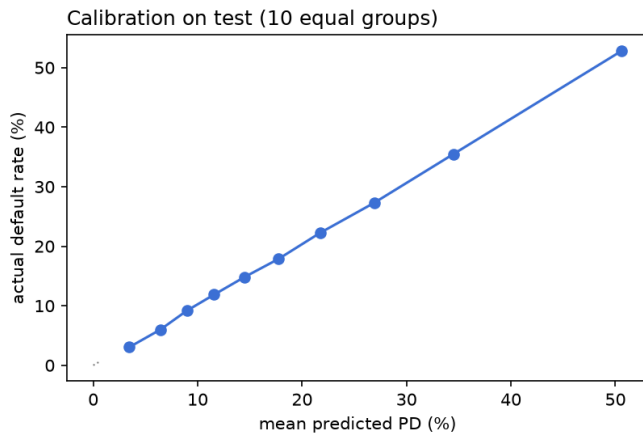
C. Pre-registered criteria

Pre-registered criterion (committed before opening the test)	Test value	Result
ROC-AUC drop vs validation <= 0.02	-0.0114	PASS
calibration gap within +/-2 pts	-0.43 pts	PASS
36-month gap within +/-2 pts	+0.22 pts	PASS
60-month gap within +/-2 pts	-2.51 pts	MISS
risk bands: default rate rises band to band	3.0% → 7.4% → ... → 52.9%	PASS
approved book default rate <= 13%	11.60%	PASS
expected / realised loss in 0.85-1.15	0.886	PASS
raw CSV -> engine matches (5,000 loans)	0.0e+00	PASS

D. Calibration by loan term: the one miss

Term	Loans	Mean PD	Actual default	Gap	ROC-AUC
36 months	162,236	14.9%	14.7%	+0.22 pts	0.7057
60 months	50,565	34.7%	37.2%	-2.51 pts	0.6933

60-month loans default **2.5 points** more often than predicted (validation: about 1 point). The ranking within 60-month loans is fine; only the level is low. The rule says nothing changes after the test, so v1.0.0 ships as is and the gap is **documented**. The remedy for a v1.1 would be a term-specific calibrator fitted on newer data.



E. Decisions and losses on test

Decision	Share (validation)	Share (test)	Default (validation)	Default (test)	Expected ÷ real loss	
APPROVE	64.7%	66.8%	11.9%	11.6%	0.97	
REVIEW	9.8%	9.3%	24.6%	25.1%	0.89	
DECLINE	25.5%	23.8%	40.1%	41.8%	0.84	
Policy on the test loans		Approved	Default rate	Real loss	Loss per \$ lent	Loss avoided
approve everyone		100.0%	20.1%	\$422.4M	13.6%	—
engine, strict (review = decline)		66.8%	11.6%	\$120.7M	6.2%	\$301.6M
engine, lenient (review = approve)		76.2%	13.3%	\$165.8M	7.4%	\$256.6M

The approve threshold chosen on validation behaves the same on test: 66.8% approved with 11.6% default (target $\leq 12\%$). Expected loss is $0.89 \times$ the real loss, close for approved loans and lower for risky ones, the same 60-month under-prediction showing through.

F. States

37 states with $\geq 1,000$ test loans: calibration gaps -3.8 to $+4.0$ points. Beyond ± 2 : AR, LA, MS, NE, NH, OK, OR. Of Stage 2's flags, LA and OR are flagged again and TN is fine. Small states move more between periods, so this supports monitoring, not a fix.

G. Stress: 2016–2018

Year (stress)	Loans	ROC-AUC
2016	293,105	0.7186
2017	169,321	0.7108
2018	56,318	0.7037

Ranking only (labels are biased toward early outcomes). It holds up 1–3 years past training.

What we learned

- The engine generalises to unseen months: ROC-AUC 0.746, overall calibration within 0.4 points, and decisions behave as designed.
- It beats LendingClub's own risk pricing at ranking these loans, without ever seeing the grade or interest rate.
- Known limitation, now confirmed on test: 60-month loans are under-predicted by about 2.5 points. It is documented, not fixed, because changing the model after seeing the test would make this test meaningless.

Next: Stage 6, the live demo (FastAPI + web page, one public link).